# Bhaskar S

Senior Full Stack Developer · Machine learning in production

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I've spent nearly nine years building Shoptree, a cloud point-of-sale, inventory and ordering platform for restaurants and retailers. I'm the lead contributor to its core platform, I built most of its ordering apps, and I've shipped machine learning to production since 2021. I also build fast on my own: six native open-source Mac apps in October 2026, made with Claude Code. Next, I want to work full-time on AI safety research.

| ~9 years | 4 in 5 | 6 | 671 |
|---|---|---|---|
| building Shoptree | of platform commits since 2022 | open-source Mac apps, Oct 2026 | Karnataka CET state rank |

## EXPERIENCE

### Shoptree Technologies Pvt Ltd — Jan 2018 – present

**Senior Full Stack Developer · Bengaluru**

Cloud point-of-sale, inventory and ordering platform for restaurants and retailers. Laravel (PHP), Vue 3, MySQL, Redis, AWS and Docker.

- Lead contributor to the core platform: about 4 in 5 commits since 2022 (2,075 of 2,690), and 80% in 2026.
- Built the customer and staff apps almost single-handedly: online ordering, table ordering, the waiter app and the kitchen management system (Vue 3).
- Cut running costs by about half by optimising key database queries and system workflows.
- Designed the inventory and management systems to scale and to plug cleanly into other applications.

### Machine learning in production — Dec 2021 – present

- Automated inventory with machine learning: demand forecasting and stock-level optimisation in production, with less manual work and fewer over- and under-stock situations.
- Built a chatbot with OCR that reads documents in real time to automate inventory work.
- Built churn-prediction models that flag at-risk customers for retention.

## EDUCATION

### IIT Madras — 2026 – present

BS in Data Science and Applications (online degree). Foundation level: statistics, mathematics, Python and English.

### BMS College of Engineering, Bengaluru — 2013 – 2015

Engineering coursework, four semesters. CGPA 8.5.

### National Pre-University College, Jayanagar — 2011 – 2013

Physics, chemistry, mathematics and electronics: 95%. Karnataka CET state rank 671.

## SKILLS

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|---|---|
| Software | PHP and Laravel · JavaScript, Vue 3, React and Node.js · Python · MySQL and PostgreSQL · Redis · Docker · AWS |
| Machine learning | PyTorch · TensorFlow · CNNs, transformers, GANs and VAEs · RAG · agents with Google ADK, MCP, LangChain and LangGraph |
| Native apps | C++, Swift and Rust on macOS, built with Claude Code |

## OPEN SOURCE

Native Mac apps published in October 2026, made with Claude Code. Small, fast and free.

**Usage?** github.com/bhaskar0210s/UsageQ — C++
Shows Claude Code session and weekly limits, pace and spending in the Mac menu bar.

**Mouse** github.com/bhaskar0210s/mouse — C++
Settings for the Logitech MX Master 3S over Logitech's HID++ protocol: 2–3 MB of memory, against about 180 MB for Logi Options+.

**Crumb** github.com/bhaskar0210s/crumb — Swift
Clipboard history for the Mac: about 500 KB on disk and 7 MB of memory, with signed releases and a Homebrew tap.

**Watcher** github.com/bhaskar0210s/watcher — C++
CPU, memory, GPU and sensors in one menu bar item: 200 KB on disk, about 10 MB of RAM.

**Revs** github.com/bhaskar0210s/revs — Rust
Line-by-line Git history for the Zed editor, as a 0.77 MB language server.

**Marky** github.com/bhaskar0210s/marky — Swift
A Markdown reader that typesets LaTeX math natively, without a web view.

**Revise** bhaskar0210s.github.io/revise — Web
Week-by-week revision for seven IIT Madras courses, with interactive statistics and linear-algebra labs. Works offline.

## MACHINE LEARNING & AI

**Restaurant Squad** github.com/bhaskar0210s/restaurant-agent — Agents
A multi-agent restaurant assistant (reservations, orders, payment) built with Google's Agent Development Kit and MCP.

**MicroGPT reasoning model** — PyTorch
A small transformer with chain-of-thought reasoning steps, inspired by OpenAI's o1.

**NotebookLM-style workspace** github.com/bhaskar0210s/notebooklm-clone — RAG
Chat with and summarise your own documents.

**Handwritten digit recognition** bhaskar0210s.github.io/Hand-Written-Recognition — CNN
A CNN trained in PyTorch that runs in the browser with TensorFlow.js, predicting as you draw.

**AirCursor** github.com/bhaskar0210s/air-cursor — Vision
Moves the pointer with your fingertip, using Apple Vision hand-pose tracking.

**brain-dot-md** github.com/bhaskar0210s/brain-dot-md — LLM memory
A brain-inspired, five-layer memory system for long-running LLM agents.

## ALSO

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|---|---|
| **Languages** | English, Kannada and Telugu; basic Hindi |
| **Interests** | AI safety, and how minds work: psychology and the brain |
| **Volunteering** | Helping children learn through local education initiatives |